

VRANJE, Serbia

WMO: 134890

Lat: 42.550N

Lon: 21.917E

Elev: 1424

StdP: 13.96

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	17.6	7.6	8.6	14.9	12.3	10.8	19.9	17.0	35.2	14.4	35.5	3.5	50	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	24.1	92.8	67.8	89.5	66.9	86.5	66.3	70.3	86.7	68.8	84.9	67.5	82.8	6.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	97.6	76.4	63.4	92.2	74.4	62.1	87.9	73.0	35.0	87.1	33.8	85.0	32.7	83.4	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	13.0	11.7	DB	6.4	97.5	6.2	3.2	1.9	99.9	-1.7	101.7	-5.2	103.5	-9.7	105.9	(4)
(5)			WB	5.7	73.1	6.0	2.1	1.4	74.6	-2.2	75.8	-5.5	77.0	-9.9	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.1	32.5	36.8	44.2	53.2	60.9	67.7	71.7	72.4	63.7	53.8	44.4	34.9		
	DBStd	15.44	8.12	8.33	7.77	6.82	6.04	5.86	5.19	5.36	6.39	7.23	7.66	7.71		
	HDD50	1889	542	372	208	44	3	0	0	0	1	48	199	472		
	HDD65	5004	1006	789	644	356	153	39	8	8	99	350	618	934		
	CDD50	3023	1	3	29	140	341	530	674	694	411	166	32	2		
	CDD65	663	0	0	0	2	26	119	217	237	59	3	0	0		
	CDH74	7124	0	0	1	50	317	1253	2300	2511	637	55	0	0		
	CDH80	2711	0	0	0	4	44	413	953	1105	188	4	0	0		
(14) Wind	WSAvg	4.7	4.6	5.2	5.6	5.2	4.8	4.6	4.9	4.6	4.3	4.0	4.3	4.8		
(15) Precipitation	PrecAvg	24.1	1.7	1.6	1.6	2.3	2.4	2.2	1.9	1.6	2.2	2.5	2.2	2.0		
	PrecMax	36.7	4.1	3.7	3.9	7.0	5.0	5.4	4.5	4.9	6.4	5.4	5.2	4.0		
	PrecMin	13.5	0.3	0.2	0.2	0.6	0.5	0.2	0.2	0.1	0.5	0.2	0.0	0.0		
	PrecStd	5.5	1.0	0.8	0.9	1.4	1.1	1.2	1.2	1.3	1.3	1.5	1.4	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.2	62.8	70.3	79.6	84.5	92.0	96.7	97.8	90.6	79.8	68.2	57.9		
		MCWB	48.8	50.5	53.6	59.2	63.5	69.0	68.6	68.0	65.1	61.9	56.5	50.0		
	2%	DB	50.6	57.7	65.9	74.0	80.6	88.1	92.6	93.5	84.9	74.6	63.4	53.0		
		MCWB	45.1	47.9	51.2	57.2	62.2	68.2	67.9	67.7	63.6	59.3	54.5	48.5		
	5%	DB	47.4	53.5	61.9	69.6	77.6	84.9	89.0	89.9	81.0	70.5	59.6	49.6		
		MCWB	43.5	45.7	49.3	54.2	61.2	66.9	67.3	66.4	62.8	58.3	52.6	45.8		
	10%	DB	44.0	49.8	57.9	66.0	74.2	81.4	85.7	86.5	77.0	66.9	56.4	46.2		
		MCWB	41.0	44.1	47.3	52.8	59.8	65.5	66.5	65.7	61.6	56.9	50.8	43.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	52.1	55.3	61.1	65.9	72.9	71.8	71.8	68.2	63.5	57.9	51.9		
		MCDB	54.9	60.6	66.6	75.2	79.0	86.7	87.9	90.2	83.6	75.7	65.0	55.9		
	2%	WB	46.1	48.6	52.2	58.2	63.8	70.0	70.0	69.8	65.6	61.2	55.6	48.8		
		MCDB	49.6	55.6	63.2	70.9	77.0	84.2	87.2	87.3	80.4	71.2	61.6	52.5		
	5%	WB	43.7	46.5	50.1	55.9	62.4	68.1	68.6	68.2	64.1	59.4	53.6	46.3		
		MCDB	46.8	52.6	59.6	66.7	74.8	82.1	85.1	85.0	77.5	67.9	59.1	49.0		
	10%	WB	41.2	44.2	48.2	53.9	60.9	66.4	67.4	66.8	62.5	57.6	51.4	43.7		
		MCDB	43.9	49.7	57.1	63.9	72.1	79.3	82.8	82.7	74.2	65.4	55.9	46.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	15.1	17.7	19.4	20.5	21.6	23.3	24.1	21.1	19.5	15.4	11.0		
		MCDBR	16.7	23.1	26.5	26.8	26.8	27.9	29.8	30.5	28.7	26.7	20.3	15.4		
		MCWBR	11.8	14.0	13.9	12.2	11.3	10.7	10.1	10.0	11.3	13.5	12.2	11.2		
		MCDBR	15.1	20.9	23.7	23.3	23.8	25.4	26.3	26.2	24.4	23.3	17.7	13.6		
	5% WB	MCWBR	11.0	13.0	12.9	11.4	11.0	10.8	10.1	10.1	10.7	12.5	10.9	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.364	0.411	0.472	0.461	0.443	0.445	0.462	0.436	0.396	0.363	0.335		
	taud		2.323	2.235	2.145	1.992	2.087	2.187	2.182	2.131	2.175	2.271	2.322	2.338		
	Ebn at Noon		250	260	261	253	260	265	263	253	250	246	237	238		
	Edh at Noon		27	35	43	54	50	46	45	46	41	33	27	25		
(44) All-Sky Solar Radiation	RadAvg		489	771	1098	1413	1720	1977	2050	1841	1326	931	590	416		
	RadStd		78	101	135	173	144	169	150	153	109	121	64	70		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.24	N/A	N/A	+1.78	+0.99	+1.06	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)		+0.97	N/A	N/A	N/A	+0.68	+0.70	N/A	-277	N/A	N/A

Nomenclature: See separate page